

Enterprise RAG API Reference

Documentation for Frontend Developers

This document serves as the standalone reference manual for frontend engineers integrating with the Enterprise RAG Assistant backend API.

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1. Overview & Base Configuration

Base URLs

- **Local Backend API:** <http://127.0.0.1:8000/api/v1>
- **Local Vite Frontend:** <http://localhost:5173>
- **Swagger / OpenAPI Documentation:** <http://localhost:8000/docs>

Global Headers

For all authenticated endpoints, you must include the JWT token obtained from the `/auth/login` endpoint:

Authorization: Bearer <jwt_access_token>

For JSON payload endpoints:

Content-Type: application/json

2. Data Types & Enums

Enums

Department (string)

Role-scoped access control boundary.

- "engineering"
- "hr"
- "operations"
- "product_support"

Category (string)

Classification of enterprise documents.

- "policy"
- "guide"
- "faq"
- "incident"
- "release_notes"

RetrievalMode (string)

Search execution strategy.

- "vector" — Standard dense vector semantic retrieval.
- "hybrid" — Dense vector search followed by CrossEncoder reranking.

ChunkingStrategy (string)

Strategy applied during document parsing.

- "fixed" — Standard fixed character chunking (512 chars, 64 overlap).
- "semantic" — Paragraph and sentence boundary-aware splitting.

Confidence (string)

Confidence metric assigned to LLM responses based on search scores.

- "high" — Top vector match score ≥ 0.5 .
- "low" — Top vector match score < 0.5 .
- "not_found" — No relevant documents found or context missing.

Role (string)

User role assigned via JWT payload.

- "admin"
- "engineering"
- "hr"
- "operations"
- "support"

3. Authentication API

POST /auth/login

Authenticates user credentials and issues a scoped JSON Web Token (JWT).

Request

- **Endpoint:** /api/v1/auth/login
- **Method:** POST
- **Content-Type:** application/json

```
{  
  "username": "bob_eng",  
  "password": "eng123"  
}
```

Response (200 OK)

```
{  
  "access_token": "eyJhbGciOiJIUzI1Ni...",  
  "token_type": "bearer",  
  "role": "engineering",  
  "departments_allowed": [  
    "engineering"  
  ]  
}
```

Errors

- **401 Unauthorized:** Invalid username or password.
- **422 Unprocessable Entity:** Invalid request body payload.

4. Chat API

POST /chat

Sends a query to the RAG system, searches scoped vector indices, reranks context, and returns an answer with citations.

Request

- **Endpoint:** /api/v1/chat

- **Method:** POST
- **Headers:** Authorization: Bearer <token>
- **Content-Type:** application/json

```
{
  "query": "What is the policy for remote work equipment reimbursement?",
  "filters": {
    "department": "hr",
    "category": "policy"
  },
  "retrieval_mode": "hybrid",
  "session_id": "a1b2c3d4-e5f6-7a8b-9c0d-112233445566"
}
```

Field Descriptions

- *query (string, required)*: User's question.
- *filters (object, optional)*: Search filters (department, category).
- *retrieval_mode (string, optional)*: "vector" or "hybrid". Defaults to "hybrid".
- *session_id (string, optional)*: UUID tracking the conversation. If omitted, the server generates a new one.

Response (200 OK)

```
{
  "answer": "Employees are eligible for a one-time remote work setup allowance of $500...",
  "sources": [
    {
      "doc_id": "8f3d1b2a-1234-5678-90ab-cdef12345678",
      "doc_name": "HR Equipment Policy v2.pdf",
      "department": "hr",
      "chunk_text": "Employees are eligible for a one-time remote work setup allowance of $500 upon starting...",
      "chunk_id": "chunk_99823",
      "score": 0.884,
      "page": 2
    }
  ],
  "retrieval_mode_used": "hybrid",
  "confidence": "high",
  "session_id": "a1b2c3d4-e5f6-7a8b-9c0d-112233445566"
}
```

Notes

- `sources[].chunk_text` is automatically truncated to 200 characters for optimal frontend card rendering.
- If confidence is "not_found", sources will be an empty array.

Errors

- **401 Unauthorized:** Missing or expired bearer token.
- **403 Forbidden:** Requested department is outside the user's `departments_allowed` list.
- **422 Unprocessable Entity:** Payload schema failure.

5. Document Ingestion API

POST /ingest

Uploads and indexes a document in Qdrant Cloud. Restricted to admin accounts.

Request

- **Endpoint:** `/api/v1/ingest`
- **Method:** POST
- **Headers:** Authorization: Bearer <admin_token>
- **Content-Type:** multipart/form-data

Body Parameters

- `file`: File blob (PDF or UTF-8 Plain Text .txt).
- `metadata`: JSON string representing document attributes.

```
{  
  "department": "hr",  
  "category": "policy",  
  "version": "2.0",  
  "doc_date": "2026-01-10",  
  "chunking_strategy": "fixed"  
}
```

Response (202 Accepted)

```
{  
  "job_id": "00112233-4455-6677-8899-aabbccddeeff",  
  "status": "processing",  
  "doc_id": "8f3d1b2a-1234-5678-90ab-cdef12345678"  
}
```

Errors

- **401 Unauthorized:** Invalid token.
- **403 Forbidden:** User is not an admin.
- **400 Bad Request:** Unsupported file format or unreadable plain text encoding.

GET /documents

Lists documents available in the registry, filtered by the caller's allowed departments.

Request

- **Endpoint:** /api/v1/documents
- **Method:** GET
- **Headers:** Authorization: Bearer <token>
- **Query Parameters:**
 - department (*string, optional*)
 - category (*string, optional*)
 - page (*integer, default: 1*)
 - page_size (*integer, default: 20*)

Response (200 OK)

```
{
  "documents": [
    {
      "doc_id": "8f3d1b2a-1234-5678-90ab-cdef12345678",
      "doc_name": "HR Equipment Policy v2.pdf",
      "department": "hr",
      "category": "policy",
      "version": "2.0",
      "chunk_count": 12,
      "created_at": "2026-01-10T09:00:00Z"
    }
  ],
  "total": 1,
  "page": 1
}
```

6. Feedback API

POST /feedback

Submits user feedback regarding an assistant response for auditing and metric tracking.

Request

- **Endpoint:** /api/v1/feedback
- **Method:** POST
- **Headers:** Authorization: Bearer <token>
- **Content-Type:** application/json

```
{
  "session_id": "a1b2c3d4-e5f6-7a8b-9c0d-112233445566",
  "query": "What is the policy for remote work equipment reimbursement?",
  "helpful": true,
  "comment": "Accurate response."
}
```

Response (200 OK)

```
{
  "status": "recorded"
}
```

7. Error Handling & HTTP Status Codes

The API uses standardized HTTP status codes and returns error responses in a unified format:

```
{
  "detail": "Error description message"
}
```

|

| **HTTP Code** | **Description** | **Causes** |

| **200 OK** | Success | Request succeeded. |

| **202 Accepted** | Job Queued | Ingestion pipeline started successfully. |

| **400 Bad Request** | Bad Input | Invalid file structure or non-UTF8 text. |

| **401 Unauthorized** | Auth Failed | Missing, invalid, or expired JWT token. |

| **403 Forbidden** | RBAC Violation | Department access breach or non-admin attempting upload. |

| **422 Unprocessable Entity** | Schema Validation Failure | Required body/query fields missing or formatted incorrectly. |

| **500 Internal Server Error** | Server Error | Unhandled backend exception or Qdrant/Groq service failure. |

8. Frontend Integration Instructions

Disabling Mock Mode for Live Backend Integration

The codebase includes a local mock API interceptor enabled by default for offline UI development. To direct frontend requests to the live backend server:

1. Open frontend/src/main.tsx.
2. Locate line 3:
`import './api/mockSetup';`
3. Comment out or remove this import line:
`// import './api/mockSetup';`
4. Verify VITE_API_BASE_URL in frontend/.env:
`VITE_API_BASE_URL=http://127.0.0.1:8000/api/v1`

Demo Accounts for Testing

The database is auto-seeded on startup with the following test accounts:

| **Username** | **Password** | **Role** | **Department Scope** |

| admin | admin123 | admin | All Departments |

| alice_hr | hr123 | hr | hr |

| bob_eng | eng123 | engineering | engineering |

| olivia_ops | ops123 | operations | operations |

| sam_support | support123 | support | product_support |